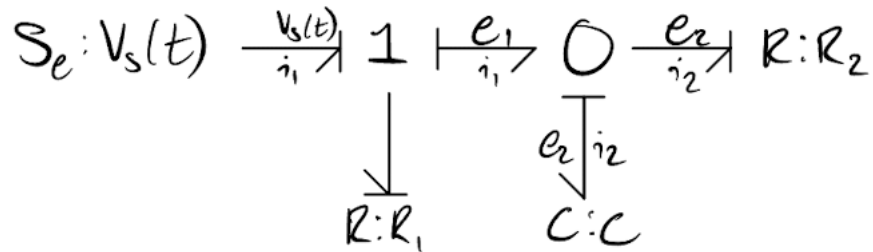


Homework #5 Solutions

9.4

a)



$$\frac{dq_c}{dt} \equiv i_c = i_1 - i_2 = \frac{e_1}{R_1} - \frac{e_2}{R_2} = \frac{V_s - e_c}{R_1} - \frac{e_c}{R_2} = \frac{V_s}{R_1} - \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{q_c}{C} = \frac{V_s}{R_1} - \frac{R_1 + R_2}{R_1 R_2} \frac{q_c}{C}$$

Or

$$\frac{CR_1 R_2}{R_1 + R_2} \dot{q}_c + q_c = \frac{CR_2}{R_1 + R_2} V_s$$

b)

$$\tau = \frac{CR_1 R_2}{R_1 + R_2}$$

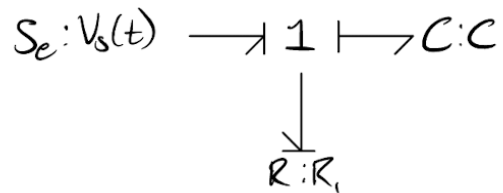
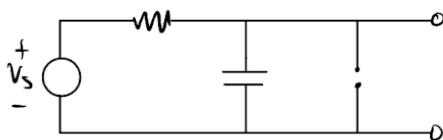
For $R_1 = R_2 = R$:

$$\tau = \frac{CR}{2} = \frac{(10^{-6} F)(10 \Omega)}{2} = 5 * 10^{-6} s$$

c)

$$\tau = \lim_{(R_2 \rightarrow \infty)} \left(\frac{CR_1 R_2}{R_1 + R_2} \right) = \lim_{R_2 \rightarrow \infty} \frac{CR_1}{\frac{R_1}{R_2} + 1} = CR_1$$

The circuit reduces to



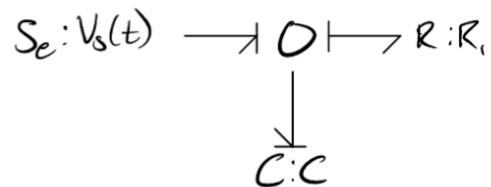
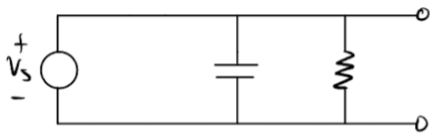
Governed by

$$CR_1 \dot{q}_c + q_c = CV_s$$

d)

$$\tau = \lim_{R_1 \rightarrow 0} \frac{CR_1 R_2}{R_1 + R_2} = \lim_{R_1 \rightarrow 0} CR_2 / (1 + \frac{R_2}{R_1}) = 0$$

The circuit reduces to

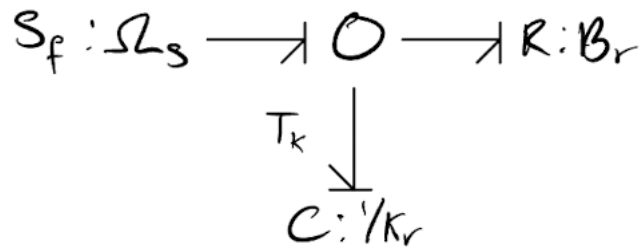


Governed by

$$q_c = CV_s \left[= \lim_{R_1 \rightarrow 0} \frac{CR_2 V_s}{R_1 + R_2} \right]$$

9.6

c)



$$\frac{d\theta_K}{dt} = \Omega_K = \Omega_s - \Omega_B = \Omega_s - \frac{K_r \Omega_K}{B_r}$$

$$\frac{B_r}{K_r} \dot{\Omega}_K + \Omega_K = \frac{B_r}{K_r} \Omega_s(t)$$

Desired variable:

$$T_K = K_r \theta_K$$

Therefore

$$\frac{B_r}{K_r} \dot{T}_K + T_K = B_r \Omega_s(t)$$

Step (or zero-state) response:

$$T_K(t) = B_r (\Omega_s)_o \left[1 - e^{-\frac{K_r t}{B_r}} \right]$$

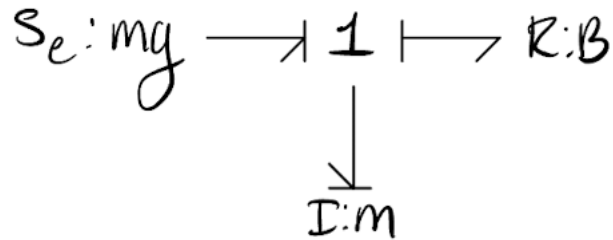
Find value: $B_r (\Omega_s)_o$

$$t = 4\tau = \frac{4B_r}{K_r}$$

9.12

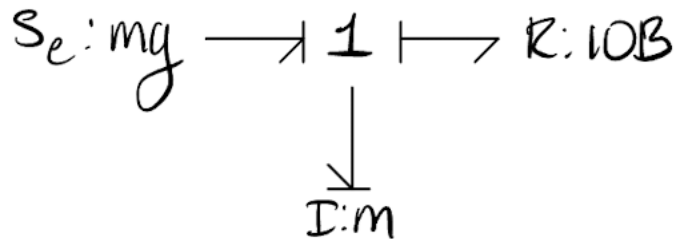
a)

w/ small chute



$$\frac{m}{B} \dot{V} + V = \frac{mg}{B}$$

w/ both (small & large) chute

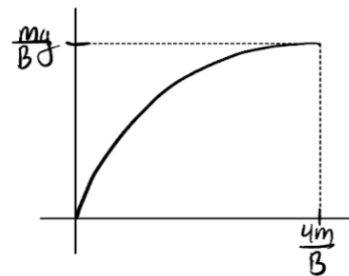


$$\frac{m}{10B} \dot{V} + V = \frac{mg}{10B}$$

b)

For $0 < t < \frac{4m}{B}$:

$$v(t) = \frac{mg}{B} \left[1 - e^{-\frac{Bt}{m}} \right]$$



@ $t \sim \frac{4m}{B}$, $V \simeq \frac{.98mg}{B}$ when large chute is deployed & changes the system's dynamics (time constant)

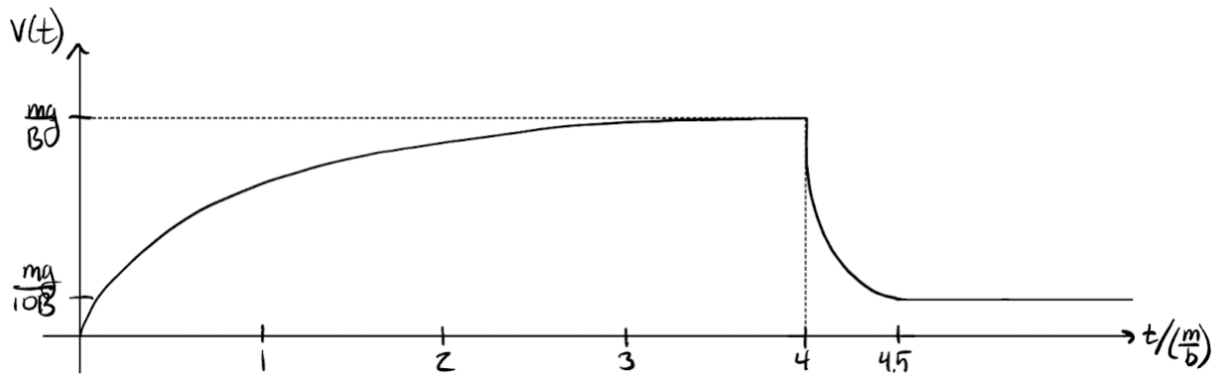
The subsequent response (i.e. $t > \frac{4m}{B}$) needs to incorporate initial velocity (energy) in the system:

For $t > \frac{4m}{B}$:

$$V(t) = \frac{mg}{B} e^{-\frac{10B\hat{t}}{m}} + \frac{mg}{10B} \left[1 - e^{-\frac{10B\hat{t}}{m}} \right] = \frac{mg}{10B} + \frac{9mg}{10B} e^{-\frac{10B\hat{t}}{m}}, \quad \hat{t} = t - \frac{4m}{B}$$

Terminal velocity, $\frac{mg}{10B}$, reached @ $t = \frac{4m}{B} + \frac{4m}{10B} = \frac{4.4m}{B}$

c)



Force on sky diver:

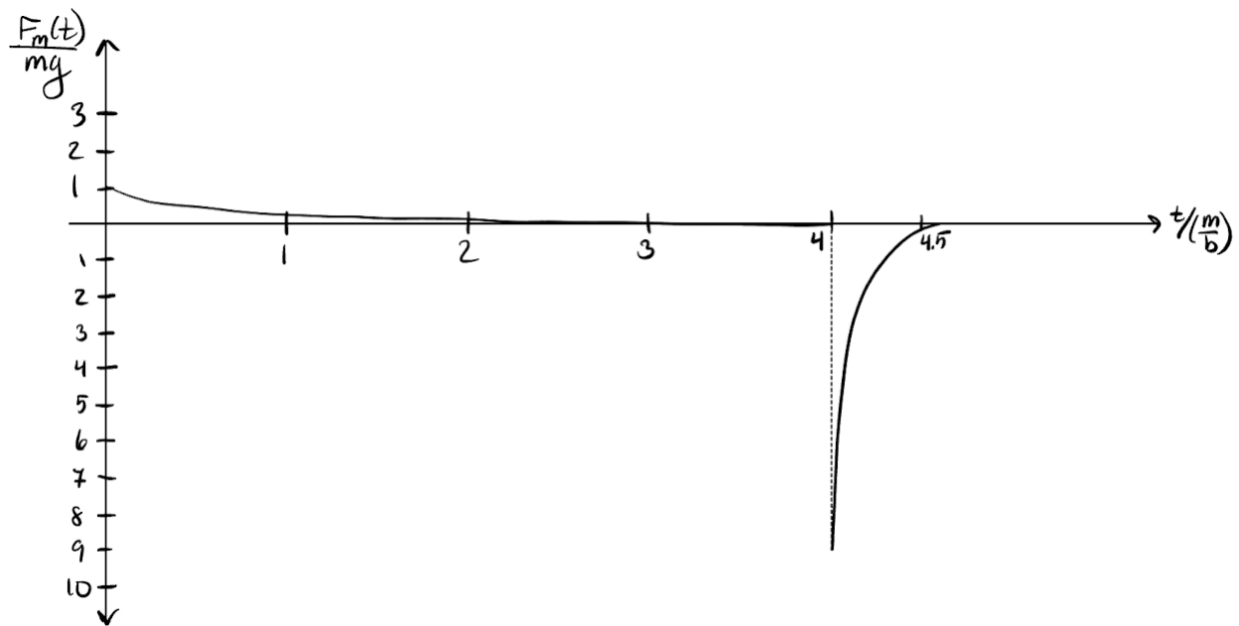
$$F_m = \dot{p} = m\dot{v}(t)$$

For $0 < t < \frac{4m}{B}$:

$$F_m = m \left(\frac{mg}{B} \right) \left[\frac{B}{m} e^{-\frac{Bt}{m}} \right] = mge^{-\frac{t}{\tau_1}}, \quad \tau_1 = \frac{m}{B}$$

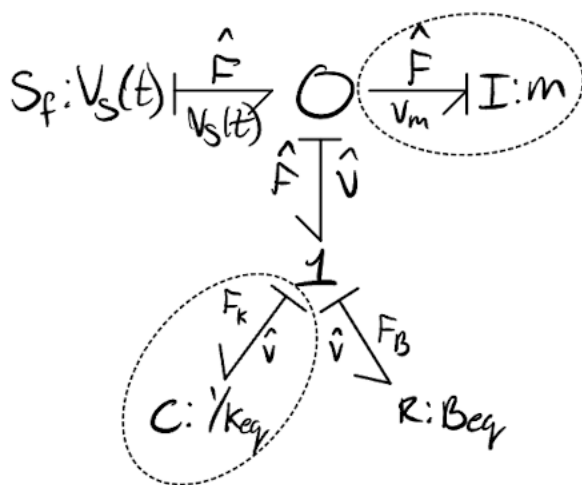
For $t > \frac{4m}{B}$:

$$F_m = m \left(\frac{9mg}{10B} \right) \left[-\frac{10B}{m} e^{-\frac{10B\hat{t}}{m}} \right] = -9mge^{-\frac{\hat{t}}{\tau_2}}, \quad \tau_2 = \frac{m}{10B}$$



9.23

a)



$$\frac{dp}{dt} \equiv \hat{F} = F_K + F_B = K_{eq}X_K + B_{eq} \left(V_s(t) - \frac{P}{M} \right)$$

$$\frac{dx}{dt} \equiv \hat{V} = V_s(t) - \frac{P}{M}$$

$$\begin{bmatrix} \dot{P} \\ \dot{X} \end{bmatrix} = \begin{bmatrix} -\frac{B_{eq}}{M} & K_{eq} \\ -\frac{1}{M} & 0 \end{bmatrix} \begin{bmatrix} P \\ X \end{bmatrix} + \begin{bmatrix} B_{eq} \\ 1 \end{bmatrix} V_s(t)$$

In terms of P:

$$P'' + \frac{B_{eq}}{M} P' + \frac{K_{eq}}{M} P = B_{eq} \dot{V}_s(t) + K_{eq} V_s(t)$$

Or $P = mV_m$:

$$V_m'' + \frac{B_{eq}}{M} V_m' + \frac{K_{eq}}{M} V_m = \frac{B_{eq}}{M} \dot{V}_s(t) + \frac{K_{eq}}{M} V_s(t)$$

$$\omega_n = \sqrt{\frac{K_{eq}}{M}} = \sqrt{\frac{16000 \frac{N}{m}}{320 \text{ kg}}} = 5\sqrt{2} \text{ Hz} = 7.07 \text{ Hz}$$

$$\eta = \frac{B_{eq}}{2\sqrt{K_{eq}M}} = \frac{1200 \frac{Ns}{m}}{2\sqrt{(4 * 400 * 320) \frac{N}{mkg}}} = 0.265$$

This system is underdamped.

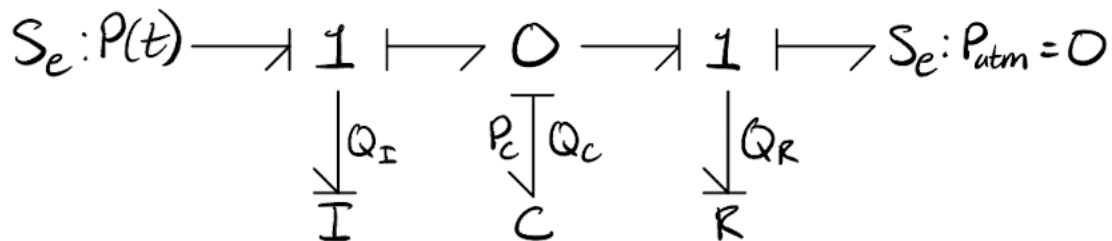
b)

To prevent oscillatory response, choose $\eta \geq 1$

$$\eta_{min} = 1 = \frac{4B_{min}}{2\sqrt{K_{eq}M}} = \frac{4B_{min}}{2\sqrt{4KM}} \rightarrow B_{min} = \sqrt{KM} = \sqrt{4000 * 320} \frac{Ns}{m} = 800\sqrt{2} \left(\frac{Ns}{m} \right)$$

9.25

a)



$$\frac{d\Gamma}{dt} \equiv P_I = P(t) - P_C = P(t) - \frac{U_C}{C}$$

$$\frac{dU_C}{dt} \equiv Q_C = Q_I - Q_R = \frac{\Gamma}{I} - \frac{P_C - P_{atm}}{R} = \frac{\Gamma}{I} - \frac{U_C}{CR}$$

$$\begin{bmatrix} \dot{\Gamma} \\ \dot{U}_C \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{C} \\ \frac{1}{I} & -\frac{1}{CR} \end{bmatrix} \begin{bmatrix} \Gamma \\ U_C \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} P(t)$$

b)

$$U_C'' + \frac{1}{CR} U_C' + \frac{1}{CR} U_C = \frac{P(t)}{I}$$

Or in terms of CP_c :

$$P_c'' + \frac{1}{CR}P_c' + \frac{1}{CI}P_c = \frac{P(t)}{CI}$$

$$\omega_n = \frac{1}{\sqrt{CI}}$$

$$\eta = \frac{1}{2R} \sqrt{\frac{I}{C}}$$

c)

Since the system is forced by a step, it's possible to prevent overshoot by choosing $\eta \geq 1$

The height of the fluid in the tank is governed by

$$h_c'' + \frac{1}{CR}h_c' + \frac{h_c}{CI} = \frac{P(t)}{IA}$$

$$\eta \geq 1, \eta = \frac{1}{2R} \sqrt{\frac{I}{C}} \geq 1$$

Therefore

$$R \leq \frac{1}{2} \sqrt{\frac{I}{C}}$$